

Optical Access Network Evolution and Transformation Enabled by AI

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Abstract. Artificial intelligence (AI) has emerged as a key enabler to address challenges such as capacity demand, dynamic resource allocation, and intelligent management through data-drive optimization and adaptive control in Optical Access Network (OAN). This paper reviews major aspects and discusses future directions for AI-driven OAN.

1. Introduction

With the rapid development of the digital economy and communication technologies, emerging services such as large-scale AI model training and industrial precision control have grown significantly. This imposes stringent requirements on the optical access network (OAN), particularly in the "last mile". These services demand ultra-high bandwidth, deterministic low latency, high reliability and dynamic resource scheduling, requirements that traditional access networks struggle to meet. Meanwhile, the maturity of AI technologies has transformed from supportive tools to key enablers of network intelligence.

ITU-T just published a technical paper on ION-2030 framework, targeting 6G and AI-driven networks, and it is actively promoted [1]. Based on this framework, two supplements, G.sup.aiBB [2] and G.sup.aiHome [3], have been initiated to collect use cases and requirements for PON and home networking.

Despite increasing service demands and ongoing standardization efforts, current access and home networks continue to face critical challenges. Inflexible resource allocation leads to low utilization, uncertain latency fails to meet service requirements, and heavy reliance on manual operations results in inefficient O&M. To address these challenges, this paper introduces an AI-driven access network architecture that integrates AI capabilities into the access network, as shown in Figure 1. A distributed multi-agent collaborative framework is proposed. In the architecture, specialized AI Agents are deployed according to the functional roles of network nodes, enabling hierarchical intelligent control and efficient service provisioning.

A network autonomy agent is deployed at the network management layer to conduct real-time network monitoring, intelligent fault diagnosis and localization, automated recovery, and intelligent scheduling of O&M tasks. At the OLT (Optical Line Terminal), an intelligent agent serves as the central control hub, responsible for global

management and coordination functions, such as dynamic resource scheduling (e.g., bandwidth, computing, and storage), collaborative optimization of network operating parameters, and unified control of multi-terminal access.

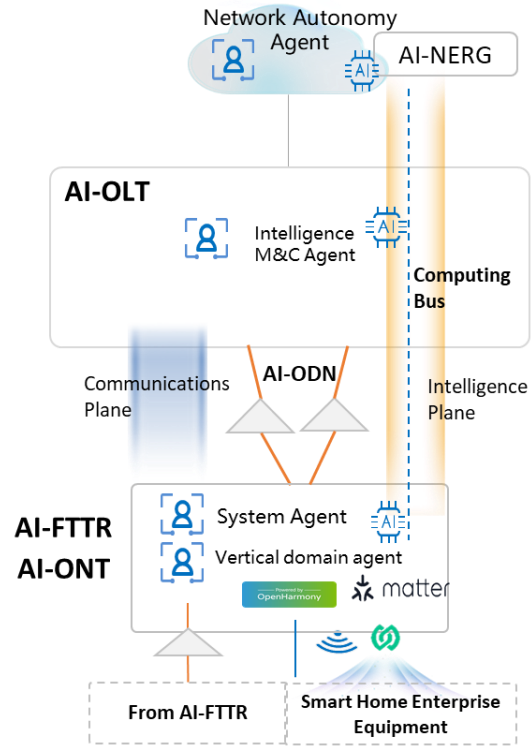


Figure 1 AI-driven access network architecture

At the FTTR (Fiber to the Room) or ONT (Optical Network Terminal) side, home/enterprise agents are deployed at the network edge. These agents enable fast local processing of user service requests, reducing reliance on cloud transmission and thereby meeting stringent low-latency and real-time service requirements.

To enhance coordination and information sharing among agents, all AI agents are interconnected through a computing power bus built on the optical link. Leveraging the high-bandwidth, low-latency of optical links, this bus supports efficient exchange of data and control signals. It enables real-time interaction, collaborative

decision-making, and coordinated task execution.

2. Evolution of AI in Optical Access Network

2.1 AI-native network as a service

Traditional OANs primarily provide CPU-based general-purpose connectivity and computing capabilities, which are insufficient to meet the end-to-end collaborative requirements of AI applications. The concept of an AI-native network introduces a “computing bus” architecture. It transforms the network from a passive pipe to an active, intelligent computing platform that enables collaboration across homes, communities, and edge clouds. This platform could provide a deterministic network foundation for interconnected intelligent agents with higher reliability.

The key advancements of this evolution are reflected in three aspects: (1) the service paradigm shifts from “best-effort” delivery to “deterministic guarantee” by efficient DBA (Dynamic Bandwidth Allocation) scheduling. (2) the service scope expands from basic home connectivity to a smart home service hub. (3) data security and privacy are inherently supported through edge-device collaborative inference. Sensitive data remains within the home domain, and only anonymized features or intermediate results are transmitted to the edge.

Key applications include AI-enabled OLT computing and AI-enabled FTTR smart home hubs. The OLT evolves into a community-level edge computing node for real-time AI inference. In the latter, diverse smart devices, wearables, and service robots are coordinated to deliver personalized and proactive services.

2.2 Integrated multimodal communication, computing, and sensing

Wi-Fi or Ethernet are the most widely adopted technologies in premises. However, they are limited to communication and lack native sensing capabilities. By using them, it is difficult to support the ubiquitous access and spatial sensing for massive, heterogeneous, low-power devices. The concept of integrated multimodal communication, computing, and sensing aims to overcome these limitations with deeply integrating multiple short-range communication technologies, such as NearLink, Bluetooth, and passive RFID into OAN. This forms a unified “optical + wireless” architecture.

The transformative value of this approach is as follows: (1) Integrated wired-wireless networking with precise synchronization, providing deterministic end-to-end experience for AI2X services,

while mitigating interference, retransmissions, and latency uncertainty; (2) a unified interconnection hub that expands the ecosystem, supporting seamless connectivity for massive smart devices across various vendors and protocols; (3) support for passive IoT and asset visibility, achieved through the integration of networks and passive RFID technologies, enabling real-time management and tracking of physical assets.

Key applications include deterministic PON and short-range communication converged networking. This ensures sub-millisecond synchronization and reliable transmission among devices. A unified smart home ecosystem centered on FTTR is another application. It acts as a control hub for managing whole-home devices and enables value-added services such as scene automation, energy optimization, and elderly care monitoring. In addition, home or campus networks can integrate RFID-based asset management systems. These systems support scenarios such as warehousing, retail, and healthcare, enabling real-time inventory tracking, location monitoring, and anti-theft protection.

2.3 Intelligent experience differentiation

In PON network, the traditional OLT allocates bandwidth using statically configured scheduling parameters. This approach cannot respond well to bursty and diverse service traffic. To address these challenges, the AI-OLT solution leverages AI to enable differentiated experience assurance that is guaranteed, operable, and measurable.

The core of AI-OLT lies in data-driven intelligent scheduling. AI-based decision-making is embedded into both the data plane and control plane. A lightweight AI inference engine is integrated into the OLT. It performs online learning using multi-dimensional data, such as historical bandwidth demand, queue backlog, delay jitter at the MAC layer, and application-level service characteristics. This enables accurate traffic prediction. Based on techniques such as deep reinforcement learning and neural networks, the AI-OLT generates optimal DBA policies.

For experience assurance, the AI-OLT establishes a closed-loop control system. This system includes perception, decision-making, execution, and feedback. First, it identifies service types in real time using flow-level features. It then maps services to experience indicators, such as latency sensitivity or high throughput. This creates a full-stack association between user experience, QoS, and physical resources. Second, for services with strict requirements, such as industrial control, the AI-OLT proactively reserves resources. It uses reinforcement learning to act before congestion occurs. This ensures deterministic low latency

and minimal packet loss. Finally, the AI-OLT opens up network capabilities through intent-based interfaces. This allows operators to define and enforce policies based on SLA requirements.

2.4 Multi-dimensional security and reliability

In PON systems, OLT is the aggregation device. It directly determines overall network survivability. Conventional protection switching mechanisms have limitations on response speed, configuration flexibility, and potential failure awareness. To address these issues, the proposed AI-OLT architecture is based on native Type B/C protection. The architecture uses the dual-homing protection with combo PON optical modules on both the OLT and ONU sides. It also integrates an AI engine for real-time situational awareness of the protection group status.

The AI-OLT continuously monitors optical layer performance. Key metrics include optical power, dispersion pre-compensation, and forward error correction (FEC) bit error rate. Monitoring is performed on both working and protection paths within milliseconds. The system uses built-in machine learning models to detect performance degradation trends and potential faults.

When failure is detected, the system reacts immediately. It leverages fast bridging and switching in the native protocol stack. Service switching is completed within tens of milliseconds to ensure no service interruption. This architecture shifts protection from passive response to proactive resilience. It significantly improves reliability and fault tolerance.

2.5 Proactive network autonomy

Proactive network autonomy is a key objective of next-generation PON systems. It enables intelligent and automated network operation. The intelligent control agent collects real-time data through a global multi-dimensional perception module. This includes physical-layer data, such as optical power and signal-to-noise ratio. It also includes MAC layer metrics, such as bandwidth utilization. Service layer indicators, such as delay and packet loss, are monitored as well. In addition, device layer information, such as device aging status, is collected. The data is standardized and processed. It is then transmitted to the agent module to perform network virtualization and state synchronization.

AI models can predict potential faults, such as fiber degradation and module failure, in advance. The planning and decision-making module generates optimal control strategies. The execution module applies these strategies automatically. This enables early fault warning, proactive performance optimization, and self-healing within seconds. As a result, the OAM efficiency of PON

systems is significantly improved.

2.6 Coordinated power saving

Figure 2 shows an overall architecture of OAN including both the PON domain and the FTTR/O domain.

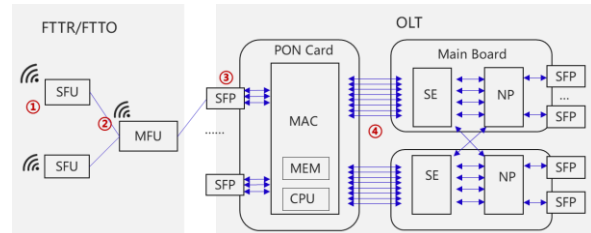


Figure 2 Coordinated power saving of OAN

AI plays a critical role in enabling this coordination. It collects multi-dimensional data, such as traffic volume, terminal status, user behavior, time periods, and service scenarios. This breaks the isolation between network elements. It also provides a global view of the network.

AI predicts traffic load and network utilization with a high accuracy. This supports proactive energy saving instead of passive sleep modes. The system can perform microsecond-level wake-up and seamless rate switching, ensuring no impact on high-priority services. AI continuously improves through online learning. It adapts to changes in user behavior, network scale, and service types. On the OLT side, it supports large-scale coordination across millions of ports, achieving overall energy savings of 40%-50%. For example, in the SFU & STA domain, AI analyzes data of both APs and STAs in real time. This includes STA activity patterns, service types (such as video, gaming), traffic load, and signal strength. Based on this data, AI generates differentiated and dynamic energy-saving strategies, realizing seamless collaboration between APs and STAs.

4. Conclusion

This paper elaborates the application of AI in OAN, including PON and home networking. AI is becoming a key direction for the evolution of OAN. It enables new capabilities such as intelligent operation and maintenance, dynamic service scheduling, fault diagnosis, and self-healing. These capabilities are for building green, efficient, and intelligent OANs. With ongoing technological advances, AI-driven OAN is also being standardized. Relevant work is progressing in ITU-T SG15 under the ION-2030 framework. These efforts aim to promote interoperability across equipment and systems. They lay a solid foundation for large-scale deployment and sustainable development of AI-OAN in the future.

References

- [1] ITU-T GSTR-ION-2030 - Technical Report on international optical networks towards 2030 and beyond, 2025
- [2] ITU-T G.Sup.ION-aiBB - Use cases and requirements of AI enhanced ION-2030 for broadband
- [3] ITU-T G.Sup.ION-aiHome - AI-enhanced home networking